

RESEARCH

Optimization of Multi-depot Semi-open Cold-Chain Logistics Paths Under Time-Dependent Road Networks

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Abstract

To address the inefficiencies and high loss rates in urban cold-chain logistics, this study investigates a complex distribution scenario characterized by a multi-depot semi-open structure, time-dependent road networks, and simultaneous pickup and delivery. A comprehensive mathematical model is developed to minimize total costs, incorporating cold-chain product deterioration, vehicle energy consumption, and customer time windows. To solve this combinatorial optimization problem effectively, a hybrid metaheuristic algorithm (FA–LNS) is proposed, which integrates the global exploration capability of the Firefly Algorithm (FA) with the local refinement strength of Large Neighborhood Search (LNS). The FA component is adapted with a discrete position update strategy, while LNS performs solution-space intensification through targeted destruction and repair operations. Extensive experiments, including benchmark comparisons with exact methods and existing heuristics, demonstrate the superior performance of FA–LNS in terms of solution quality and computational efficiency. Furthermore, sensitivity analyses are conducted on key logistics parameters—vehicle departure times, dynamic driving speeds, and return strategies—highlighting the importance of modeling time-varying congestion, semi-open routing flexibility, and departure-time optimization. The findings offer actionable managerial insights to support strategic decision-making in cold-chain logistics.

Keywords Multi-depot · Time-dependent road networks · Semi-open · Vehicle routing problem · Cold-chain logistics

1 Introduction

In recent years, market demand for various cold-chain products has increased significantly. These include meat, fruits, vegetables, aquatic products, dairy products, frozen foods, vaccines, biological preparations, pharmaceuticals, and other temperature-sensitive items. This rapid expansion is primarily driven by population growth, increasing consumer preferences for fresh and high-quality goods, and heightened awareness of the importance of strict temperature control to maintain product integrity. Moreover, advancements in global supply chains, e-commerce platforms, and logistics technologies have facilitated more efficient transportation and distribution of these perishable products. As a result, the food, healthcare, and pharmaceutical industries increasingly rely on cold-chain logistics to ensure product safety, quality, and efficacy, thereby further driving market growth [1].

Unlike traditional logistics, cold-chain logistics requires seamless integration across multiple operational links, along with the adoption of cutting-edge technologies and management practices to ensure the smooth functioning

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of the entire supply chain. The success of cold-chain logistics heavily depends on precise temperature control, efficient tracking systems, and robust coordination among various stakeholders. As third-party logistics providers expand rapidly in both quantity and capability, the traditional model of relying on regional single-distribution-center systems has become increasingly inadequate in meeting the diverse and dynamic needs of modern logistics. This has resulted in a gradual shift toward more flexible and scalable distribution systems, such as open and semi-open multi-point distribution networks, which can handle a higher volume of goods, reduce bottlenecks, and provide more responsive services to meet customer demands in real-time [2]. However, the implementation of these systems introduces significant challenges, including high loss rates, frequent “broken chain” risks [3], low distribution efficiency, and cost control difficulties. These challenges compromise both the sustainable development of cold-chain logistics and food safety assurance. Within the complex urban distribution environment [4, 5], cold-chain orders exhibit characteristics such as small batch sizes, high delivery frequency, and diverse demand patterns [6, 7], imposing stringent requirements—particularly for timeliness and accuracy—on distribution services. This demand heterogeneity necessitates urgent optimization of distribution routes through comprehensive approaches incorporating order profiles, traffic dynamics, cargo attributes, and customer requirements for resource efficiency maximization.

Beyond inherent urban distribution complexities, cold-chain logistics faces environmental uncertainties, among which urban traffic congestion represents a critical challenge. Congestion amplifies delivery time unpredictability and induces delays that jeopardize product freshness, quality, and safety [8], affecting perishables including food, pharmaceuticals, and vaccines [9]. Such disruptions impair supply chain operations and customer satisfaction, where delayed/damaged deliveries degrade both product integrity and consumer experience. Consequently, trust erosion occurs, potentially diminishing market reputation, operational efficacy, and sector competitiveness long-term.

Consequently, optimizing urban cold-chain distribution routes demand integrated consideration of customer heterogeneity, innovative distribution paradigms, and multidimensional factors including order profiles, traffic patterns, and meteorological conditions. This study addresses the optimization of multi-depot semi-open cold-chain routes in time-varying road networks to simultaneously improve operational efficiency and service effectiveness. We propose to resolve complex distribution scenarios through synergistic integration of advanced logistics information systems and computational optimization techniques. This framework establishes an adaptive routing model that simultaneously enhances delivery efficiency, reduces product loss rates, and ensures operational stability and reliability.

The remainder of this paper is structured as follows. Section 2 comprehensively reviews relevant literature, establishing the research necessity and significance. Section 3 formalizes the problem and develops a mathematical model for multi-depot semi-open cold-chain logistics under time-varying networks. Section 4 introduces an innovative hybrid algorithm integrating the Firefly Algorithm (FA) and Large Neighborhood Search (LNS) for route optimization. Section 5 presents sensitivity analysis of algorithm parameters along with comparative experiments across different problem scales, including exact versus heuristic algorithm benchmarks. Section 6 details computational experiments with real-world instance data, results analysis, and three additional sensitivity studies. Sections 7 and 8 synthesize key findings, address research limitations, propose future directions, and derive managerial implications from multi-stakeholder perspectives. This research ultimately delivers scientifically-validated solutions for enhancing cold-chain distribution performance in practical urban traffic environments.

2 Literature Review

2.1 Related Studies

Regarding Time-Dependent Vehicle Routing Problems (TDVRP), Jin et al. developed a mathematical model for real-time Cash-in-Transit (CIT) routing problems under traffic conditions. The goal of this model is to

comprehensively consider economic and environmental factors, and a nearest neighbor search algorithm was designed to solve the problem [10]. Huang et al. constructed a versatile routing model for the TDVRP; they proposed an approximation method to generate improved solutions under stochastic traffic scenarios [11]. Ma et al. focused on the TDVRP within the context of shared autonomous electric vehicle services, with a particular emphasis on optimizing departure times and speeds; their objective was to reduce costs by considering factors such as travel distance, energy consumption, and time [12]. Luo et al. addressed the Time-Dependent Green Vehicle Routing Problem (TDGVRP) amid traffic congestion, aiming to devise vehicle scheduling strategies to reduce carbon emissions. To this end, they constructed a set partitioning model and developed a branch-price-and-cut algorithm to generate efficient solutions [13]. Franco et al. focused on the bi-objective TDVRP, considering the probabilities of delivery failures, and aimed to reduce both operational costs and delivery failure rates. To address this complex challenge, they formulated a Mixed-Integer Linear Programming model [14]. Wang et al. addressed the dynamic TDGVRP in waste collection, aiming to improve transportation efficiency for construction waste and reduce carbon emissions. They formulated a 0–1 programming model and designed a two-stage algorithm to solve the problem [15]. Li et al. investigated the Time-Dependent Vehicle Routing Problem with Time Windows (TDVRPTW) and spatio-temporal distance, aiming to reduce travel time while enhancing customer satisfaction. To achieve these objectives, they established relevant models and developed an enhanced multi-objective evolutionary algorithm to generate efficient solutions [16]. Liu et al. devised an Adaptive Large Neighborhood Search (ALNS) algorithm for TDGVRP to reduce carbon emissions [17]. Said et al. corrected errors in the computer implementation of the branch-and-price algorithm for the TDGVRP [18]. Castellucci et al. created an exact approach using logic-based Benders decomposition to address TDVRP challenges in congested urban operations [19]. Hou et al. optimized urban cold-chain routing under carbon constraints with a hybrid genetic algorithm, where their model integrated demand, service time, speed, and load factors. Empirical validation provided actionable insights for logistics optimization [20].

Due to fierce competition in logistics and distribution markets and growing environmental concerns, the vehicle routing problem for co-distribution has attracted significant research attention. Multi-depot co-distribution has more advantages over traditional distribution models. Dubey et al. focused on surplus food recovery and redistribution for food banks via a front-end model, developing a mixed-integer programming model for the Multi-Depot Vehicle Routing Problem with Time Windows (MDVRPTW), split pickup, and split delivery [21]. Liu et al. focused on MDVRPTW to minimize travel costs, developing a Mixed-Integer Linear Programming model and a search algorithm with a perturbation mechanism [22]. Shi et al. focused on balancing economic and environmental impacts, aiming to minimize operating costs and carbon emissions. They developed a bi-objective optimization model and employed the NSGA-III algorithm to obtain the Pareto optimal solution set [23]. Stodola et al. tackled the Multi-Depot Vehicle Routing Problem with Drones (MDVRP-D), where trucks and Unmanned Aerial Vehicles (UAVs) are used for delivery, with the primary objective of enhancing customer service efficiency. They formulated a relevant mathematical model and introduced a metaheuristic algorithm based on Ant Colony Optimization (ACO) [24]. Liu et al. addressed the MDVRPTW of simultaneous delivery and extraction faced by manufacturing companies, aiming to minimize total travel cost via a Mixed-Integer Linear Programming model and a variable neighborhood search algorithm [25]. Su et al. focused on optimizing the Multi-Depot Green Vehicle Routing Problem with Time Windows (MDGVRPTW) by considering customer satisfaction, proposing a lightweight genetic algorithm integrated with variable neighborhood search [26]. Jolfaei et al. focused on the MDVRPTW to minimize total travel time, presenting a mathematical model and a hybrid LNS algorithm for solving large-scale instances [27]. Wang et al. addressed multi-warehouse pickup-and-delivery vehicle routing with time windows and dynamic demand in urban logistics recycling, developing a dual-objective mathematical model and an adaptive NSGA-II algorithm [28]. Xu et al. investigated operational planning and profit-sharing integration in cooperative collection-transport systems under the circular economy, examining three scenarios: non-cooperative, centralized, and decentralized. For routing problems, they proposed two heuristics and a dynamic programming approach [29]. Lim et al. addressed prevalent local optima in complex MDVRPTW (hindering

optimal solution search). To overcome this, they proposed the SW-MopGA framework, which introduced a controller for population diversity maintenance and a two-step approach to enhance solution quality [30]. Giovanni et al. addressed the two-tier multi-depot vehicle routing problem (incorporating robot stations and time windows), formulating a Mixed-Integer Linear Programming approach and a Multi-Start Iterated Local Search metaheuristic (enhanced with crossover and repair operators) to efficiently identify optimal solutions [31]. Wang et al. improved route planning and resource allocation flexibility for MDVRPTW via a hybrid heuristic algorithm [32].

2.2 Research Analysis

While previous research on TDVRP and MDVRPTW has laid a solid foundation, the integration of these two dimensions—time-dependent routing and MDVRPTW—remains underexplored, especially in the context of cold-chain logistics. Most existing studies either focus on static traffic assumptions or neglect the operational flexibility required in multi-depot scenarios. However, for cold-chain distribution, time sensitivity, perishability, and energy consumption create a tightly coupled decision environment that necessitates more comprehensive modeling.

To address these gaps, this study proposes a novel urban cold-chain logistics routing model that jointly considers the characteristics of time-varying road networks and simultaneous pickup-and-delivery operations. The model operates under a “multi-depot semi-open structure”, where vehicles are allowed to return to the nearest depot after deliveries, enhancing route flexibility and reducing travel distances.

In addition to modeling complexity, algorithmic innovation is also essential. To efficiently solve this problem, we design a “hybrid metaheuristic algorithm (FA-LNS)” that combines the global search capability of the Firefly Algorithm (FA) with the local refinement power of Large Neighborhood Search (LNS). The FA component simulates phototactic behavior to rapidly generate diverse initial solutions. To adapt it to discrete routing problems, we tailor a novel position update mechanism suited to permutation-based encoding. The LNS component then performs “destruction and repair” operations to refine route quality, selectively removing and reinserting customer nodes to reduce cost. Moreover, we introduce a “departure time optimization strategy” to mitigate the impact of traffic congestion and reduce in-transit time and cooling energy consumption. This strategy is validated through sensitivity analysis, revealing its significant effect on total delivery cost and product deterioration. Together, the proposed model and algorithm offer an integrated approach for solving real-world urban cold-chain routing problems under dynamic and complex conditions.

3 Problem Formulation

3.1 Problem Description

The problem studied in this paper is formulated as follows: for multiple distribution centers and several customer points (each with known coordinates and clear pickup-and-delivery demands), homogeneous cold-chain vehicles are used to provide services. To improve operational efficiency, the cargo compartments of vehicles are divided into two separate zones: one for loading delivery goods and the other for collecting pickup goods. The operational process is as follows: vehicles depart from a distribution center, load delivery goods for the corresponding route, synchronously complete delivery and pickup tasks according to the demands of each customer point along the way, and finally return to the nearest distribution center. The problem must incorporate the time-dependent characteristics of urban road traffic (vehicle speed varies with time), and vehicles must comply with weight constraints throughout the process. The research objective is to determine a delivery plan that meets all constraints and minimizes the total delivery cost.

To streamline the model construction while accounting for the complexities of real-world delivery operations, the following assumptions are adopted:

- (1) The vehicles' departure times are set based on the time window of the first customer on the delivery route. After completing the delivery tasks, the vehicles return to the nearest distribution center.
- (2) For each customer, the location, pickup demand, delivery demand, time window, and service duration are predetermined and constant. Customer demands are indivisible and can only be fulfilled by a single logistics vehicle.
- (3) Urban traffic congestion is assumed to be influenced only by time, with no consideration given to specific locations or areas.

3.2 Symbols and Interpretations

The symbols and interpretations are shown in Table 1.

3.3 Calculation of Travel Time

To better reflect the vehicle's speed in accordance with the actual traffic conditions, this section draws upon previous research and utilizes a triangular function $v(t) = \alpha \sin(\beta t) + \gamma$ to model the vehicle's speed, where α , β , γ represent different road parameters [33], and the specific method for calculating the vehicle's travel time is as follows: first of all, to determine the departure time lt_{ik} from point i , in the $[lt_{ik}, lt_{ik} + t_{ij}]$ time speed time enclosed in the area of the distance traveled in the time period, can be calculated by the following equation:

Table 1 Symbols and interpretations

Symbols	Interpretations
N_*	Set of customer points: $N = \{1, 2, \dots, n\}$
M_*	Set of distribution centers: $M = \{1, 2, \dots, m\}$
K_*	Set of vehicles: $K = \{k \mid k = 1, 2, \dots, K\}$
$[E, L]$	Operating time window of distribution centers
$[a_i, b_i]$	Time window for customer point i
Q	Vehicle's maximum load capacity
$V_{\max}$	Vehicle usage limit
q_i	Demand at customer point i
p_i	Pickup demands at customer point i
s_i	Service time at customer point i
d_{ij}	Distance from point i to point j
Q_{ijk}	Load carried by vehicle k from point i to point j
T_k^0	Departure time of vehicle k from distribution center
T_k^{end}	Arrival time of vehicle k at distribution center
at_{ik}	Arrival time of vehicle k at point i
bs_{ik}	Service start time of vehicle k at point i
wt_{ik}	Waiting time of vehicle k at customer point i
lt_{ik}	Departure time of vehicle k from point i
x_{ijk}	Binary variable: $x_{ijk} = 1$ if vehicle k travels from point i to point j , otherwise $x_{ijk} = 0$
Y_{ik}	Binary variable: $Y_{ik} = 1$ if vehicle k visits point i , otherwise $Y_{ik} = 0$

$$d_{ij} = \int_{l_{ik}}^{l_{ik}+t_{ij}} v(t)dt \quad (1)$$

Given that the distance from point i to point j , the departure time l_{ik} from point i , and the arrival time at_{jk} at point j are known, the vehicle travel time t_{ij} can be obtained by integrating the mathematical formula with the upper limit.

3.4 Logistics Cost

3.4.1 Fixed Cost

The fixed cost C_1 includes the basic maintenance of distribution vehicles, driver wages, and other related expenses. These costs are dependent on the number of vehicles used and are calculated as follows:

$$C_1 = P_1 \sum_{k \in K_*} \sum_{i \in M_*} \sum_{j \in N_*} x_{ijk} \quad (2)$$

3.4.2 Refrigeration Cost

Since cold-chain products are perishable and require strict temperature control during distribution, refrigeration cost are incurred. These refrigeration costs are divided into three components: C_{21} is the refrigeration cost incurred during transportation, C_{22} refers to the cost during waiting, C_{23} denotes the refrigeration cost associated with the opening and closing of compartment doors. The detailed calculation of these refrigeration costs is as follows:

$$C_2 = C_{21} + C_{22} + C_{23} \quad (3)$$

$$C_{21} = P_2 \sum_{k \in K_*} \sum_{i,j \in M_* \cup N_*} t_{ijk} x_{ijk} \quad (4)$$

$$C_{22} = P_2 \sum_{k \in K_*} \sum_{i \in N_*} \max((a_i - at_{ik})Y_{ik}, 0) \quad (5)$$

$$C_{23} = P_3 \sum_{i \in N_*} \sum_{k \in K_*} Y_{ik} s_i \quad (6)$$

3.4.3 Damage Cost

Cold-chain products are highly sensitive to temperature. Exceeding a certain temperature threshold will cause a certain degree of loss. We use the continuous index loss function $D(t) = D_0 e^{-\partial t}$, where $D(t)$ is the product quality at time t , D_0 represents the product quality when departing from the distribution center, t indicates the duration of transportation undergone by the product, and $\varepsilon_1, \varepsilon_2, \varepsilon_3$ denote the spoilage rates of the product [34, 35].

The cold-chain damage costs associated with simultaneous pickup and delivery mainly include: goods damage caused by transportation duration (delivery products C_{31} , pickup products C_{32}), and goods damage C_{33} resulting from temperature rise due to air convection caused by door opening and closing during the service process. Equation (7) represents the cost of goods loss throughout the entire distribution process.

$$C_3 = C_{31} + C_{32} + C_{33} \quad (7)$$

$$C_{31} = P_4 \sum_{i \in N_*} \sum_{k \in K_*} q_i \left(1 - e^{-\varepsilon_1 (bs_{ik} - T_k^0)} \right) Y_{ik} \quad (8)$$

$$C_{32} = P_4 \sum_{i \in N_*} \sum_{k \in K_*} p_i \left(1 - e^{-\varepsilon_2 (T_k^{end} - lt_{ik})} \right) Y_{ik} \quad (9)$$

$$C_{33} = P_4 \sum_{k \in K_*} \sum_{i \in N_*} \sum_{j \in M_* \cup N_*} Q_{ijk} (1 - e^{-\varepsilon_3 s_i}) Y_{ik} \quad (10)$$

3.4.4 Fuel Cost

In this study, environmental cost is considered as a key optimization objective, primarily comprising fuel consumption calculated using the Comprehensive Emission Model (CMEM). Developed by Barth et al. in 2008, the CMEM model primarily considers the impact of load, speed, acceleration, road gradient, and other vehicle characteristic parameters on fuel consumption [36]. The fuel consumption f_{ijk} of vehicle k from customer point i to customer point j can be expressed as:

$$f_{ijk} = \omega \gamma \mu \left\{ (a_{ij} + g \sin \theta_{ij} + g C_r \cos \theta) (m_0 + Q_{ijk}) D_{ijk} + 0.5 C_d \rho A \left(v_{ijk}^h \right)^2 D_{ijk} \right\} + \omega \gamma K N V D_{ijk}^h / v_{ijk} \quad (11)$$

$$C_4 = C_e \sum_{i, j \in M_* \cup N_*} \sum_{k \in K_*} x_{ijk} \cdot f_{ijk} \quad (12)$$

3.5 Mathematical Model

$$\text{Min} C = \text{Min} (C_1 + C_2 + C_3 + C_4) \quad (13)$$

$$\sum_{k \in K_*} Y_{ik} = 1, \forall i \in N_* \quad (14)$$

$$\sum_{k \in K_*} \sum_{i \in M_*} x_{ijk} \leq 1, \forall j \in N_* \quad (15)$$

$$\sum_{j \in M_*} \sum_{k \in K_*} x_{ijk} \leq V_{max}, \forall i \in N_* \quad (16)$$

$$\sum_{i \in M_*} \sum_{j \in N_*} x_{ijk} = \sum_{i \in N_*} \sum_{j \in M_*} x_{ijk}, \forall k \in K_* \quad (17)$$

$$\sum_{i \in N_*} q_i Y_{ik} \leq Q, \forall k \in K_* \quad (18)$$

$$\sum_{i \in N_*} p_i Y_{ik} \leq Q, \forall k \in K_* \quad (19)$$

$$0 \leq Q_{ijk} \leq Q, \forall k \in K_*, \forall j, i \in N_* \quad (20)$$

$$at_{ik} + wt_{ik} = bs_{ik}, \forall k \in K_*, \forall i \in N_* \quad (21)$$

$$bs_{ik} + s_i = lt_{ik}, \forall k \in K_*, \forall i \in N_* \quad (22)$$

$$E \leq a_i \leq bs_{ik} \leq b_i \leq L, \forall k \in K_*, \forall j \in N_* \quad (23)$$

$$t_{ijk} = \xi(s_i + at_{ik}), \forall k \in K_*, \forall j, i \in N_* \quad (24)$$

$$x_{ijk}, Y_{ik} \in \{0, 1\}, \forall k \in K_*, \forall j, i \in M_* \cup N_*, \quad (25)$$

where Eq. (13) is the total cost. Constraint (14) restricts every customer to be visited only once. Constraint (15) indicates that all vehicles depart from one of the distribution centers. Constraint (16) indicates that the number of vehicles used is within the specified limit. Constraint (17) denotes that the vehicle does not have to return to the original distribution center after completing the assigned task. Constraint (18) represents the load limit of vehicle k departing from the distribution center. Constraint (19) denotes load constraints for vehicle k returning to the distribution center. Constraint (20) denotes the loading constraint during the working period of vehicle k . Constraint (21) denotes the continuity of distribution time, i.e., the sum of the arrival time of vehicle k at point i and the waiting time at point i is equal to the start of service at point i . Constraint (22) denotes the continuity of distribution time, where the sum of the start of service at point i and the length of service at point i is equal to the time of departure from point i . Constraint (23) denotes that the start of service of vehicle k to customer point i must satisfy the time window $[a_i, b_i]$ of that customer point and also satisfy the working time window $[E, L]$ of the distribution center. Constraint (24) denotes that the time t_{ijk} for vehicle k to travel from point i to point j is a decision variable, and its value is determined by the moment at_{ik} at which vehicle k arrives at point i and the service duration s_i at point i . Constraint (25) represents a binary decision variable.

4 Algorithm Design

4.1 Encoding and Decoding

This section uses real number coding. The problem contains $|M|$ customers and $|K|$ vehicles, and defines the chromosome length as $N = |M| + |K| - 1$, which is randomly arranged and stored in C_Num . Numbers greater than $|M|$ are considered as path splitting points.

The encoding and decoding processes are shown in Fig. 1. First, identify the positions where values exceed M , perform path cutting, and sequentially apply load and time window constraint tests. If the constraints are satisfied, the path is stored; otherwise, the allocation is terminated, and a new vehicle is dispatched for service until all customers are scheduled. Second, for the stored paths, extract the positions of the first and last customers, calculate their distances from the distribution centers, and assign the starting and ending distribution centers based on the shortest distance principle (the starting and ending distribution centers do not have to be the same).

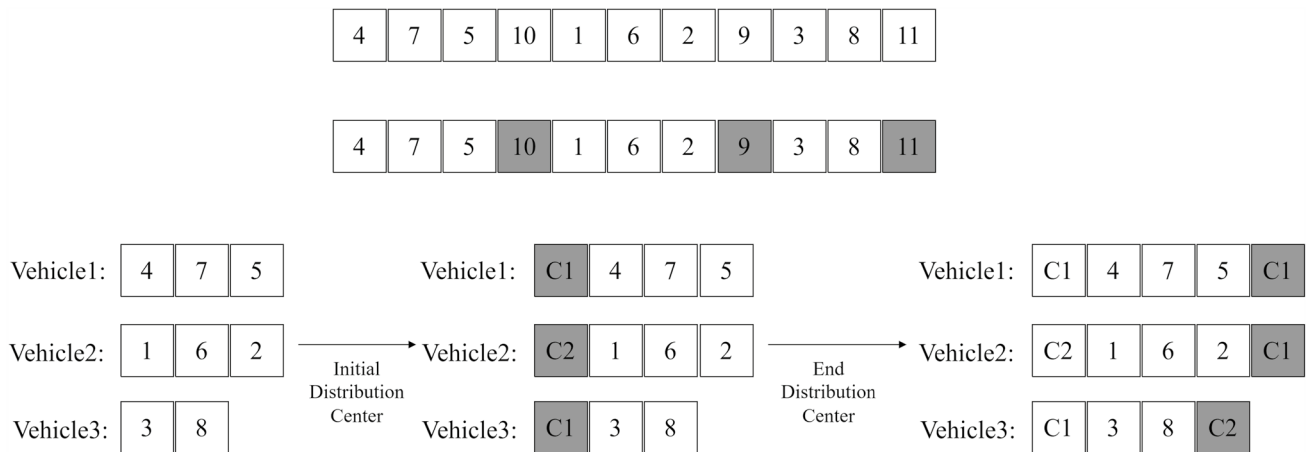


Fig. 1 Decoding schematic

4.2 Calculation of Luminous Intensity

For each individual x_i , its objective function value $f(x_i)$ is calculated, and the luminous intensity I_i is inversely proportional to the objective function value:

$$I_i = \frac{1}{f(x_i)} \quad (26)$$

4.3 Design of Discrete Distance Metric

In the standard Firefly Algorithm (FA) for continuous optimization problems, the Cartesian distance (Euclidean distance) is used to measure the similarity between individuals. It is calculated as the geometric distance between position vectors in the continuous space, which reflects the straight-line distance between two points in the physical space and serves as the core basis for continuous domain search. For discrete combinatorial optimization problems, this paper specifically designs a discrete distance measurement formula:

$$r_{ij} = \sum (\text{Individual}_i \neq \text{Individual}_j) \quad (27)$$

This formula quantifies the structural differences between discrete solutions by counting the number of differing elements at corresponding positions in two path coding sequences. It is independent of the geometric properties of the physical space and directly adapts to the coding characteristics of discrete solutions, providing a basis for the adaptive regulation of crossover operations and enabling the effective search of FA in the discrete solution space.

4.4 Update Firefly Locations

The position update of the firefly population is achieved through information interaction between individuals, and the specific process is as follows:

First, for any two individuals (fireflies), i and j in the population (denoted as Individual_i and Individual_j), the superiority of individuals is judged by their objective function values. If $f(\text{Individual}_j) < f(\text{Individual}_i)$, the learning process of individual i towards individual j is triggered. Second, the discrete distance metric r_{ij} between two individuals is calculated. Third, the crossover segment length l_{ij} is determined based on the distance r_{ij} :

$$(l_{ij} = \mathcal{U}(1, \lceil r_{ij} \cdot \gamma^{\text{gen}} \rceil), \quad (28)$$

where γ is the light absorption coefficient, gen is the current iteration number, and $\lceil \cdot \rceil$ is the ceiling function. Then the crossover operation is performed to generate new individuals:

$$((\text{Individual}'_i, \text{Individual}'_j) = \text{crossover}(\text{Individual}_i, \text{Individual}_j, l_{ij}). \quad (29)$$

Finally, the population was updated using the greedy acceptance criterion: if $f(\text{Individual}'_i) < f(\text{Individual}_i)$, then update individual i to $\text{Individual}'_i$; otherwise, keep the original individual unchanged.

The above update mechanism adapts the FA to combinatorial optimization problems by converting position movement in continuous space into cross-evolution of discrete solutions. It not only retains the core feature of the algorithm of “learning from better individuals” but also balances global exploration and local development capabilities through adaptive crossover length regulation.

4.5 Local Search

To further enhance the algorithm’s search capability, this paper designs several local optimization operators:

(1) Relevance removal operator: Randomly remove several customers from the current solution σ to form set C ; the remaining customers constitute a partial solution σ' . Randomly select a customer z from C , then sort the remaining customer points in σ' by their correlation with z in descending order. Select the first customer from this sorted list and transfer it to C . Repeat this step until a specified number of customers are selected, yielding the final set C . The correlation between customers is defined as:

$$R(i, j) = \frac{1}{d_{ij} + v_{ij}} \quad (34)$$

(2) Regret insertion operator: For customers to be reinserted, first calculate the cost increment of inserting them into each feasible route. The “regret value” is derived from the cost difference between the optimal and sub-optimal routes. Prioritize reinserting the customer with the largest regret value (if no feasible route exists, create a new route). Finally, all removed customers were reintegrated back into the routes.

4.6 Time Optimization Strategy

In vehicle path research, most studies assume that all vehicles leave the distribution center simultaneously. However, departure time is crucial for avoiding traffic congestion and reducing costs. Optimizing departure times can significantly cut fuel consumption and travel time while meeting customer needs. Therefore, this paper proposes a departure-time optimization strategy to improve distribution efficiency and reduce costs through smart scheduling.

The vehicle departs from the distribution center to the first customer point to provide pickup-and-delivery services. The departure time is determined based on the congestion time period, the time window requirement of the first customer point, the earliest departure time of the vehicle ES , the latest departure time LS , and the nearest congestion time period $[T_1, T_2]$. The optimal departure time t_{best} is calculated as shown in Eq. (30) according to the travel time formula:

$$t_{\text{best}} = \begin{cases} ES, ES \leq T_1 \text{ or } ES \geq T_2 \\ LS, T_1 < ES < LS < T_2 \\ T_2, ES < T_2 < LS \end{cases} \quad (30)$$

The optimal departure time t_{best} is then evaluated to ensure that the vehicle satisfies the initial load constraints and time window constraints, and the evaluation function is shown in Eq. (31) as follows:

$$\rho_1 \text{Load}(t_{\text{best}}) + \rho_2 TW(t_{\text{best}}) \leq 0 \quad (31)$$

where ρ_1 and ρ_2 are the penalty coefficients for violating the load constraint and the time window constraint, $\text{Load}(t_{\text{best}})$ and $TW(t_{\text{best}})$ are the values for violating the load constraint and the time window constraint at departure time t_{best} .

4.7 Algorithmic Steps

The algorithm steps are shown in Fig. 2.

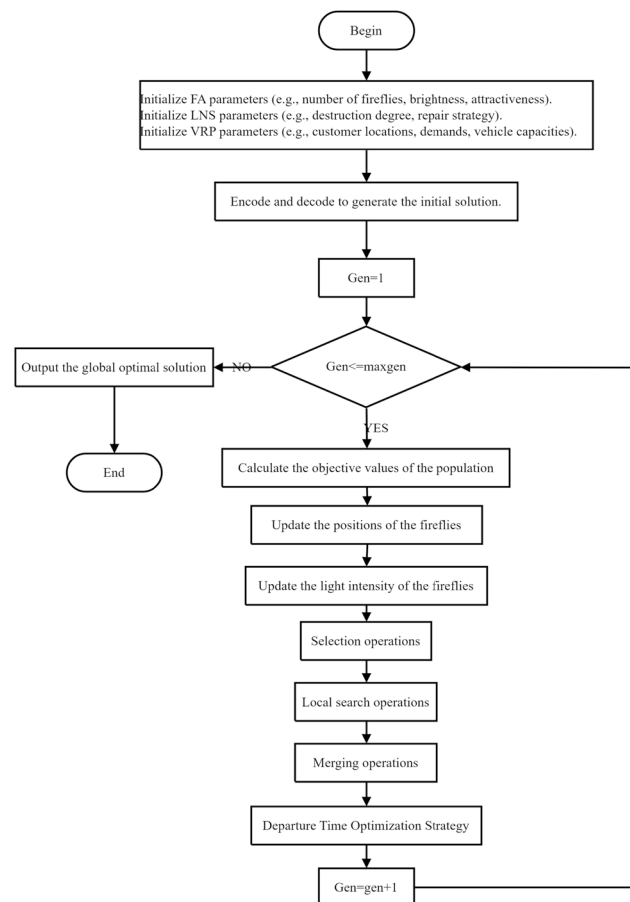
5 Parameter Sensitivity Analysis and Algorithm Comparison

The program is implemented using Matlab R2023b, and Gurobi is implemented using Python 3.10. Both run on an Intel(R) Core(TM) i7-14700HX 2.10 GHz processor (16.0 GB of RAM) under the Windows 11 operating system.

5.1 Parameter Sensitivity Analysis

This section conducts a sensitivity analysis of algorithm parameters using Solomon benchmark instances. The effects of different values on the optimization results are analyzed by setting different iteration times, light absorption coefficients, selection probabilities, and the number of removed customers, and the experiment is run ten times under each parameter setting to obtain its average value. Based on the above analysis and Fig. 3,

Fig. 2 Algorithm flow chart



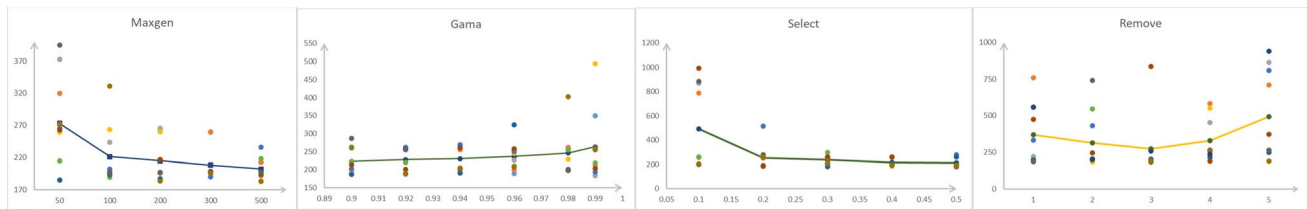


Fig. 3 Sensitivity analysis plot for each parameter

the following conclusions can be reached: (1) The maximum number of iterations is negatively correlated with the results. As the number of iterations increases, the search range expands, and the probability of the algorithm finding a better solution increases. However, at the same time, the computation time increases significantly. This suggests that a trade-off between solution quality and computational efficiency is necessary for practical applications. (2) The light attraction coefficient is positively correlated with the results. An increase in the attraction coefficient leads to a reduction in the range of step factor values, making the algorithm more likely to fall into a local optimum. This implies that the attraction coefficient must be carefully selected in algorithm design to avoid overly restricting the search space. (3) The selection probability is negatively correlated with the results. While selecting a larger population for local search operations can expand the search range and increase the probability of finding better solutions, it also significantly increases computation time. Therefore, it is essential to strike a balance between the search scope and computational efficiency. (4) The number of “removals” in the search operation should not be maximized. A reasonable choice of the number of removals helps to balance exploring the new solution space with utilizing the current solution structure, thus improving the overall performance of the algorithm.

Building on the above analysis and conclusions, the relevant algorithm parameters for subsequent experiments are configured as follows: maximum number of iterations: 300–500; light absorption coefficient: 0.95; selection probability: 0.5; number of removed customers: approximately 20% of the total number of customers.

5.2 Exact Algorithm Comparison

To validate the effectiveness and superiority of the proposed algorithm, a comparative analysis was conducted between our algorithm and the Gurobi solver using VRPTW instances from the Solomon dataset, which includes three distribution types: clustered (C), random (R), and mixed (RC). The maximum running time for the Gurobi solver was set to 3600 s, with the current optimal solution output upon exceeding this time limit. The solving results are presented in Table 2 (The one with superior performance in solution time has been highlighted in bold).

As shown in Table 2, Gurobi can rapidly find optimal solutions for small-scale problems. However, as the problem scale expands (with customer nodes increasing to 25), its solution time increases significantly, and its performance within 3600 s becomes unsatisfactory. In contrast, the FA-LNS algorithm can obtain optimal solutions within a reasonable time frame, with particularly prominent advantages in handling R-type and RC-type data. This is attributed to the relatively scattered distribution of customer nodes in R-type and RC-type data, where the FA-LNS algorithm can more efficiently plan vehicle routes, avoid traffic congestion, and reduce travel distances in such widely distributed scenarios. The results demonstrate that the proposed FA-LNS algorithm is effective in solving this problem and exhibits superiority in handling large-scale data with widely distributed customers.

5.3 Heuristic Algorithm Comparison

To verify the superiority of the proposed algorithm, it is compared with heuristic algorithms such as the Large Neighborhood Search (LNS) and Particle Swarm Optimization (PSO). Parameter settings are as follows: for PSO, inertia weight $w = 0.9$, individual learning factor $c_1 = 1.5$, and global learning factor $c_2 = 2$; for LNS, the

Table 2 Comparison results with the Gurobi solver

Instance	N^1	Gurobi			FALNS			GAP ⁴ (%)
		DIS ²	K	t/s^3	DIS ²	K	t/s^3	
C101	5	42.42	1	<1	42.42	1	3.06	0.00
	10	58.33	1	1.72	58.33	1	4.29	0.00
	25	191.81	3	47.53	191.81	3	13.23	0.00
C201	5	70.45	1	<1	70.45	1	3.33	0.00
	10	152.29	2	1.74	152.29	2	4.60	0.00
	25	215.54	2	35.12	215.54	2	13.07	0.00
R105	5	141.85	2	<1	141.85	2	3.56	0.00
	10	231.53	2	4.12	231.53	2	4.60	0.00
	25	464.67	3	711.11	464.67	3	13.49	0.00
R201	5	141.85	2	<1	141.85	2	3.58	0.00
	10	249.20	2	3.27	249.20	2	4.46	0.00
	25	463.48	4	813.03	463.48	4	13.98	0.00
RC201	5	89.13	1	<1	89.13	1	3.11	0.00
	10	183.14	2	3.30	183.14	2	4.64	0.00
	25	391.30	3	3600+	359.23	3	11.65	8.20
RC207	5	89.13	1	<1	89.13	1	3.16	0.00
	10	144.67	1	3.23	144.67	1	5.54	0.00
	25	313.98	3	3600+	298.95	3	12.64	4.79

¹ N represents the customer size

²DIS represents the travel distance

³ t/s represents the solution time

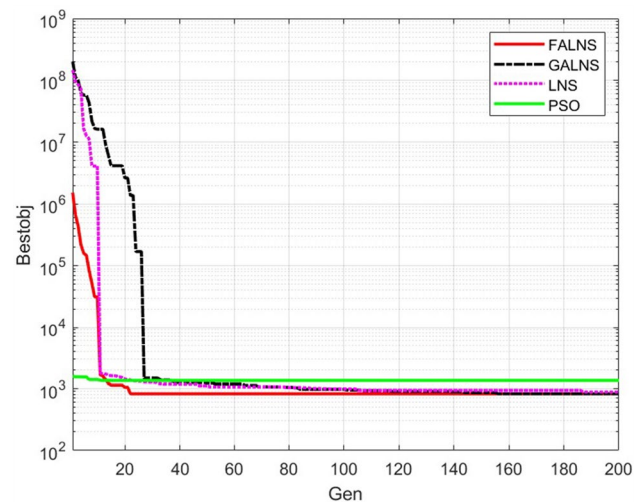
⁴GAP = $(DIS_{\text{Gurobi}} - DIS_{\text{FALNS}}) / DIS_{\text{Gurobi}}$
Table 3 Comparison results with the heuristic algorithm

Instance	N	BKS ¹	PSO		LNS		GALNS		FALNS	
			DIST	GAP ² (%)	DIST	GAP ² (%)	DIST	GAP ² (%)	DIST	GAP ² (%)
C101	50	362.40	394.52	8.85	362.40	0.00	362.40	0.00	362.40	0.00
C102	50	361.40	391.08	8.20	362.17	0.21	362.17	0.21	362.17	0.21
C103	50	361.40	486.33	34.70	381.97	5.71	391.74	8.39	361.71	0.09
RC201	50	684.80	792.24	15.69	811.97	18.57	793.05	15.81	684.80	0.00
RC202	50	653.60	739.89	12.90	699.07	6.95	680.43	4.38	653.60	0.00
RC203	50	615.60	777.05	20.96	593.60	-3.57	641.96	4.28	617.90	0.37
C101	100	828.94	828.94	0.00	828.94	0.00	828.94	0.00	828.94	0.00
C102	100	828.94	850.99	2.66	828.94	0.00	828.94	0.00	828.94	0.00
C103	100	828.94	953.14	15.01	828.96	0.00	829.56	0.08	828.96	0.00
RC201	100	1261.80	1246.84	13.08	1689.60	33.90	1440.88	14.19	1413.23	12.00
RC202	100	1092.30	1290.19	18.12	1249.30	14.37	1172.62	7.35	1102.22	0.91
RC203	100	1049.62	1212.18	15.50	1235.07	17.70	1113.42	6.10	1070.40	2.00
AVG	—	—	—	16.34	—	9.32	—	6.26	—	1.55

¹BKS represents the best-known solutions

²GAP represents the relative error between the algorithm-obtained solutions and the BKS

number of customers to be removed $D = 10$; for GALNS, crossover probability $P_c = 0.9$ and mutation probability $P_m = 0.1$. The comparison results of the algorithms are presented in Table 3. Meanwhile, taking a 100-customer

Fig. 4 Iterative comparison plot of heuristic algorithms

instance from the C101 dataset as an example, the convergence performance of each algorithm is compared, with detailed results shown in Fig. 4.

Table 3 indicates that in terms of solution quality, the optimal values obtained by the FA–LNS algorithm have a minimum error of 0%, a maximum error of 12%, and an average error of 1.55% compared with the best-known solutions (BKS). The relative error increases as the customer scale expands. The solution results of the FA–LNS algorithm are significantly superior to those of PSO, LNS, and GALNS. Through comparative analysis of the operation results, the effectiveness of FA–LNS is verified. Meanwhile, the iteration graph in Fig. 4 shows that the iterative convergence capability of the proposed algorithm is superior to that of the other three algorithms.

6 Case Study and Sensitivity Analysis

6.1 Case Data

An illustrative analysis was performed utilizing two distribution hubs and the 25 customer nodes they serve in Shenyang, 25 adjacent locations were randomly chosen as customer service points. The spatial distribution of

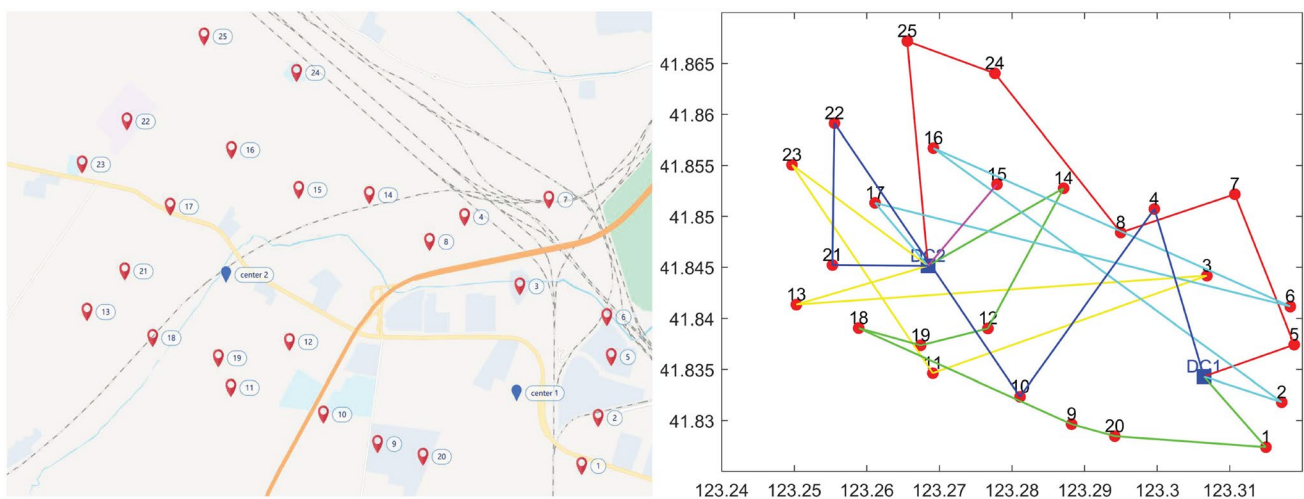
**Fig. 5** Example optimization before and after comparison

Table 4 Data case

Number	Longitude	Latitude	LTW ¹	RTW ²	ST ³	Demand	Pick demand
Center1	123.3064379	41.83430987	0	230	0	0	0
Center2	123.2684303	41.8451369	0	230	0	0	0
1	123.31714	41.83180763	50	60	10	0	73
2	123.3068385	41.84416988	116	126	10	69	0
3	123.2996033	41.85072548	149	159	10	56	0
4	123.3188549	41.83743024	34	44	10	0	85
5	123.3183274	41.84115387	99	109	10	52	0
6	123.3106763	41.85217365	81	91	10	88	0
7	123.294977	41.84841162	95	105	10	0	17
8	123.2882041	41.82964896	97	107	10	30	0
9	123.2810949	41.83236566	124	134	10	31	0
10	123.2690782	41.8342843	67	77	10	0	20
11	123.2766733	41.83902536	63	73	10	28	0
12	123.2502891	41.84133377	159	169	10	43	0
13	123.2871002	41.85277348	32	42	10	0	38
14	123.2779212	41.85314745	61	71	10	0	97
15	123.269166	41.85670806	75	85	10	0	91
16	123.2611327	41.85132767	157	167	10	8	0
17	123.2588695	41.83906235	87	97	10	0	89
18	123.2674045	41.8373661	76	86	10	36	0
19	123.2941464	41.82847386	126	136	10	11	0
20	123.2552577	41.84521374	62	72	10	0	41
21	123.255538	41.85912428	97	107	10	75	0
22	123.2496856	41.85506657	68	78	0	0	104
23	123.2776364	41.86402926	153	163	10	0	80
24	123.2655621	41.86719184	172	182	10	55	0
25	123.31714	41.83180763	50	60	10	0	73

¹LTW represents the customer's left time window

²RTW represents the customer's right time window

³ST represents the customer's service time

these customers is depicted in Fig. 5. Table 4 shows the detailed case data. Owing to the minimal variations in distances calculated from latitude and longitude coordinates, all entries in the inter-customer distance matrix were proportionally enlarged by a factor of 10,000. The algorithm parameters are consistent with those in Sect. 5.1, the vehicle speed function is set as: $v(t) = 70 \sin\left(\frac{1}{2}t\right) + 80$. The experiments are conducted with the objective of minimizing the total cost. Drawing on relevant existing studies, the definitions and values of parameters related to the model and costs are presented in Table 5.

6.2 Experimental Results and Analysis

Employing the model parameters formulated in Sect. 6.1, the aforementioned instance data are solved. The spatial distribution map of customer locations prior to solution and the vehicle delivery route map after solution are illustrated in Fig. 5, while the detailed vehicle travel information is presented in Table 6. The experimental results indicate that six refrigerated vehicles are required to complete the delivery task. The departure times for each vehicle have been optimized to some extent to avoid traffic congestion on the route. The arrival times at each customer point meet the required standards, and the return times to the distribution center also adhere to the constraints. There was no violation of this load constraint throughout the delivery process. The total cost is

Table 5 Vehicle parameter interpretation and values

Parameter	Interpretation	Values
P_1	Vehicle usage cost	200
P_2	Refrigeration cost per-unit transportation and waiting-time	0.16
P_3	Cooling cost per-unit service-time	0.25
P_4	Product price	20
ε_1	Corruption rate 1	0.001
ε_2	Corruption rate 2	0.001
ε_3	Corruption rate 3	0.002
C_e	Fuel cost factors	6.43
ω	Heat released per gram of fuel burned	0.00135
γ	Conversion factor	0.0277
μ	$\mu = 1/1000\eta \times \eta_{tf}$	0.00277
K	Engine friction coefficient	0.2
N	Engine speeds	33
V	Engine displacement	5
A	Vehicle surface area	3.912
C_r	Rolling resistance coefficient	0.01
C_d	Drag coefficient	0.7
m_0	Vehicle deadweight	3500
a_{ij}	Accelerations	0

Table 6 Results of example calculations

VN ¹	ROUT ²	T0 ³	ARR ⁴	TT ⁵	WT ⁶	BT ⁷
1	DC1 → 5 → 7 → 8 → 24 → 25 → DC2	35	40.92 → 56.85 → 92.38 → 106.10 → 165.30	18.35	80.36	188.26
2	DC2 → 23 → 11 → 3 → 13 → DC2	69	75.14 → 81.28 → 93.81 → 129.29	21.07	51.9	173.89
3	DC2 → 14 → 12 → 19 → 18 → 9 → 20 → 1 → DC1	39.69	41.07 → 52.45 → 74.37 → 88.30 → 100.17 → 14.29 → 138.22	24.42	46.67	178.82
4	DC1 → 2 → 16 → 6 → 17 → DC2	51	54.66 → 68.32 → 88.99 → 115.21	22.22	58.48	167.67
5	DC2 → 21 → 22 → 10 → 4 → DC1	63	64.24 → 75.48 → 108.91 → 141.64	22.45	43.97	165.1
6	DC2 → 15 → DC2	62	63.59	3.79	0	75.79

¹ VN represents the vehicle number² ROUT represents the vehicle routing³ T0 represents the departure time of each vehicle⁴ ARR represents the moment of reaching each customer⁵ TT represents the travel time of each vehicle⁶ WT represents the wait time of each vehicle⁷ BT represents the back time of each vehicle

1312.2122, which is broken down into the following components: fixed costs of 1200, fuel costs of 17.38, refrigeration costs of 64.85, and cargo damage costs of 29.98.

6.3 Sensitivity Analysis

6.3.1 Sensitivity Analysis of Departure Time

To validate the efficacy of the proposed time optimization strategy, this section presents a comparative analysis of two distinct departure paradigms: (1) the “SAME” paradigm, where all vehicles depart simultaneously

Table 7 Sensitivity analysis of departure time

CASE	VN	ROUT	T0	TT ¹	OBJ	DC ²	FC ³	RC ⁴
SAME	1	DC2 → 23 → 22 → 24 → 25 → DC2	0	967.01	2499.46	1205.83	11.82	81.81
	2	DC2 → 15 → 6 → 1 → DC	0					
	3	DC1 → 2 → 19 → 8 → 3 → 4 → 17 → DC2	0					
	4	DC1 → 5 → 7 → 9 → 10 → 20 → DC1	0					
	5	DC2 → 21 → 11 → 18 → DC2	0					
	6	DC2 → 14 → 12 → 16 → 13 → DC2	0					
DIFF	1	DC1 → 2 → 11 → 7 → 18 → 1 → DC1	51	505.41	1295.82	13.57	21.69	60.57
	2	DC2 → 22 → 6 → 20 → 4 → 24 → 25 → DC2	94					
	3	DC2 → 15 → 12 → 8 → 3 → 10 → 17 → DC2	62					
	4	DC1 → 5 → 21 → 16 → DC2	54.7					
	5	DC2 → 23 → 19 → 9 → 13 → DC2	69					
	6	DC2 → 14 → DC2	52.59					

¹TT' represents the total working times for all vehicles

²DC represents the damage cost

³FC represents the fule cost

⁴RC represents the refrigeration cost

Table 8 Sensitivity analysis of vehicle speed

V	VN	TT'	OBJ	FC	DC
V1	5	457.94	1206.89	28.09	117.45
V2	6	458.02	1299.51	31.83	7.06
V3	6	565.42	1483.98	33.39	189.20

at time zero from arbitrary distribution centers, and (2) the “DIFF” paradigm, where vehicles strategically select departure times based on delivery task requirements. Under the DIFF framework, departure times are dynamically optimized to avoid traffic congestion while strictly adhering to customer time window constraints. The comparative results of these two operational strategies are summarized in Table 7.

As demonstrated by the results in Table 7, while both departure paradigms (SAME and DIFF) utilize an identical number of vehicles, significant differences emerge in operational efficiency and cost performance. Specifically, the DIFF paradigm exhibits a substantial reduction in travel time (TT') and damage cost (DC) compared to the SAME paradigm. This observation suggests that adopting dynamically scheduled departure times effectively minimizes vehicle en-route duration and mitigates perishable product deterioration in cold-chain logistics. In addition, although the fuel cost (FC) increased slightly, the DIFF departure strategy showed a large reduction in total cost of delivery (OBJ) and cooling cost (RC). Together, these findings show that time-optimized departure strategies can reduce costs and increase efficiency and sustainability in logistics distribution by reducing transportation time, product loss, and total distribution costs.

6.3.2 Sensitivity Analysis of Vehicle Speed

To verify the impact of vehicle driving speed on distribution costs, this study conducts comparisons under different congestion states, sets up six groups of control experiments with different scales, and takes the minimum total cost as the objective. The comparison results are shown in Table 8. V1 (with small, gentle speed fluctuations and a long cycle): the vehicle speed is $v(t) = 50 \sin\left(\frac{1}{3}t\right) + 70$; V2 (this experiment): the vehicle

speed is $v(t) = 70 \sin\left(\frac{1}{2}t\right) + 80$; V3 (with large, abrupt speed fluctuations and a short cycle): the vehicle speed is $v(t) = 80 \sin\left(\frac{3}{4}t\right) + 90$.

As demonstrated in Table 8, significant disparities emerge among different speed function scenarios. Specifically, when the speed function exhibits small, gentle fluctuations with a long cycle (V1), vehicle utilization increases, accompanied by reductions in total cost, fuel cost, and travel time. Conversely, under a speed function with large, abrupt fluctuations and a short cycle (V3), total cost, fuel cost, and travel time increase. These findings highlight the substantial impact of speed function volatility on cold-chain logistics distribution. Notably, the cargo damage cost under our proposed scenario (V2) is the lowest among all speed function designs. This suggests that optimizing speed functions through dynamic routing strategies—which adapt to real-time traffic conditions to avoid congestion and minimize in-transit time—can effectively reduce distribution costs. Furthermore, these results empirically underscore the necessity of incorporating time-varying road network dynamics in cold-chain logistics modeling.

6.3.3 Sensitivity Analysis of Return Strategy

To verify the effectiveness of the “semi-open” return strategy in the model, it is compared with the “non-open” return strategy (returning to the original distribution center after delivery). The results are shown in Table 9.

As shown in Table 9, compared to the “semi-open” distribution mode, the “non-open” configuration demonstrated significant inferiority across three key metrics: travel distance, damage cost, and total cost. These experimental findings underscore that adopting the “semi-open” return strategy—where vehicles return to the nearest distribution center upon completion of deliveries—effectively mitigates product spoilage, reduces operational costs, and minimizes travel distance. This outcome not only validates the operational efficiency of proximity-based return protocols but also emphasizes the critical importance of flexible routing strategies in optimizing cold-chain logistics performance.

7 Managerial Insights

This study yields several data-driven managerial implications that can inform operational decision-making in urban cold-chain logistics:

1. **Adopt dynamic departure scheduling to minimize product loss and enhance efficiency:** Sensitivity analysis (Table 7) reveals that the DIFF departure strategy significantly reduces travel time and product damage

Table 9 Sensitivity analysis of return strategy

CASE	VN	Route	DIST	DC	OBJ
NON-OPEN	1	DC2 → 21 → 16 → 3 → 13 → DC2	8631	34.92	1321
	2	DC1 → 7 → 8 → 6 → 10 → 4 → 17 → DC1			
	3	DC2 → 23 → 12 → 9 → 20 → 24 → 1 → DC2			
	4	DC1 → 2 → 22 → 25 → DC1			
	5	DC2 → 14 → 5 → 11 → 19 → 18 → DC2			
	6	DC2 → 15 → DC2			
SEMI-OPEN	1	DC2 → 14 → 2 → 16 → 17 → DC2	7546	29.98	1312
	2	DC2 → 21 → 19 → 18 → 22 → 13 → DC2			
	3	DC2 → 23 → 8 → 3 → 25 → DC2			
	4	DC2 → 12 → 11 → 6 → 20 → 4 → 1 → DC1			
	5	DC1 → 5 → 7 → 9 → 10 → 24 → DC2			
	6	DC2 → 15 → DC2			

costs, despite a marginal increase in fuel expenditure. Cold-chain operators are, therefore, encouraged to adopt flexible, time-optimized departure schedules rather than fixed departure schemes. This approach mitigates en-route delays and refrigeration load, ultimately improving both sustainability and customer satisfaction.

2. **Leverage real-time traffic and speed profiles for adaptive routing:** Our results (Table 8) demonstrate that route efficiency is highly sensitive to traffic speed variability. High-frequency, short-period speed functions lead to more efficient vehicle utilization and reduced distribution costs. Consequently, logistics planners should incorporate real-time traffic data and predictive speed models into their routing systems to avoid congestion and optimize transit timing. This is especially critical in urban areas with fluctuating traffic patterns.
3. **Implement semi-open routing for operational flexibility and cost reduction:** As shown in Table 9, the semi-open return strategy—where vehicles return to the nearest depot instead of a fixed home base—results in lower travel distances, reduced product damage, and decreased overall costs. This finding suggests that proximity-based return policies can significantly enhance route efficiency and are particularly beneficial in multi-depot scenarios.
4. **Exploit multi-depot configurations to optimize network-wide efficiency:** The case studies underscore the operational advantages of multi-depot semi-open logistics structures. These configurations reduce the need for cross-region transfers, shorten last-mile distances, and enable higher vehicle reuse rates. Cold-chain enterprises operating in metropolitan areas with dispersed demand nodes should evaluate the cost-effectiveness of establishing regional satellite depots.
5. **Integrate algorithmic route optimization into daily planning:** The demonstrated superiority of the FA–LNS algorithm across multiple performance metrics indicates the practical value of integrating metaheuristic optimization into daily route planning. Logistics decision-makers should consider embedding such algorithms into their fleet management platforms, enabling real-time re-routing and efficient response to operational disruptions.

Together, these findings translate technical insights into actionable strategies, equipping cold-chain logistics managers with evidence-based guidance to improve distribution efficiency, sustainability, and resilience.

8 Conclusion

8.1 Discovery

This study investigates the urban cold-chain logistics distribution problem under a multi-depot semi-open structure, incorporating simultaneous pickup and delivery, product deterioration, and time-dependent road networks. A comprehensive cost-minimization model was developed that explicitly accounts for transportation, refrigeration, fuel, and cargo damage costs. To efficiently solve this complex problem, we proposed a hybrid FA–LNS heuristic, which was validated against both exact and heuristic benchmarks. Extensive case experiments and sensitivity analyses on vehicle departure time, driving speed functions, and return strategies confirm the importance of dynamic scheduling and route flexibility. Specifically, optimized departure times significantly reduce total cost and cargo loss, dynamic speed functions enhance routing adaptability to traffic fluctuations, and semi-open return strategies outperform non-open configurations in minimizing distribution cost and spoilage risk. These findings not only validate the practicality of our model and algorithm but also offer actionable guidance for improving the efficiency and sustainability of urban cold-chain logistics systems.

8.2 Limitations and Future Research Directions

Despite the promising results, several limitations warrant further exploration. First, the formulation of damage cost, although refined in this study to better reflect the dynamics of simultaneous pickup and delivery, may still be simplified compared to real-time temperature fluctuations and cargo-specific decay rates. Second, while we conducted sensitivity

analyses on key operational variables, real-world uncertainties such as traffic accidents, vehicle breakdowns, or sudden weather changes were not explicitly modeled. Future research can extend this work by integrating stochastic variables and real-time IoT data to enhance the responsiveness and robustness of cold-chain routing strategies. In addition, multi-objective optimization frameworks that simultaneously consider service quality, carbon emissions, and equity among depots could offer a more holistic foundation for green and resilient cold-chain logistics design.

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Data availability The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declaration

Conflict of interest The authors report there are no competing interests to declare.

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